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Customer churn prediction for fixed wireless access: The case of a regional internet service provider

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A B S T R A C T

Customer churn remains a significant area of concern in the telecommunications sector due to its high prevalence. As Fixed Wireless Access (FWA) gains traction, the field is witnessing heightened competition, especially with the integration of 5G technology, which serves dual purposes for cellular communications and FWA. This research delves into a comprehensive analysis of risk factors that impact churn via Random Survival Forests, the Cox regression model, and Kaplan-Meier Survival Plots. Our results indicate that fiber, sales zone (i.e., brand presence), age, speed, and discounts are some of the more important variables with regard to the cumulative hazard ratio. Although discounts were effective in many sales zones, their impact was diminished for more expensive internet services. The insights garnered provide a foundational basis for telecom providers to devise robust strategies to curtail customer churn over time, ensuring sustained growth and customer satisfaction in the burgeoning FWA market.

1. Introduction

The broadband landscape in the U.S. has witnessed a profound transformation in recent years. Notably, the monthly data usage per household for broadband has seen a significant uptick, jumping from 270 GB in 2018 to 641 GB in 2023 (OpenVault, 2023). This trend underlines the increasing dependency of U.S. households and businesses on the internet, necessitated by data-intensive tasks such as streaming services, online shopping, remote work, and e-learning endeavors.

In addition to the surge in data consumption, the strides in speed benchmarks are equally compelling. At the time of this writing, median download speeds for fixed broadband in U.S. households were 207 Mbps, and median upload speeds were 23 Mbps (Ookla, 2023). This far outpaces the Federal Communications Commission's (FCC) long-standing definition of "broadband," which is pegged at a constant connection delivering a minimum of 25 Mbps for downloads and 3 Mbps for uploads (Cooper, 2018). More recently, Broadband Equity, Access, and Development (BEAD) funding requires a minimum of 100 and 20 Mbps for download and uploads respectively (National Telecommunications and Information Administration, 2024). Current consumer demand in internet usage has spurred increases in internet speed, which is reflective of technological advancements and industry/government funding.

Rural residents, comprising 20.0% of the U.S. population (US Census Bureau, 2022), often encounter a less satisfactory digital experience compared to their urban counterparts. The digital disparity in these rural areas primarily arises due to the financial challenges of covering sparsely populated areas. Among the various connectivity solutions available, including satellite, DSL, cable, fiber optics, and fixed wireless, the U.S. government's Rural Digital Opportunity Fund (RDOF) predominantly supports the latter two

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(Dawson, 2022), whereas BEAD funding primarily supports fiber optics.

Numerous technologies aimed at serving rural regions have been largely sidelined due to various challenges that limit their ability to serve those regions effectively. DSL, once a mainstream technology, is now perceived as a dated wired solution. Cable has seen some advances with regard to throughput; however, fiber optics is now preferred for the creation of the infrastructure backbone (Kaur et al., 2023). That being said, new landlines are somewhat impractical/expensive to connect directly to rural households. For example, the FCC has estimated that $80 billion would be needed to implement a point-to-point fiber solution, but only $20.4 billion is available in the Rural Digital Opportunity Fund (Campbell, Castro, & Wessel, D., 2021). The BEAD Program was established to address some of this funding gap; however, the program has experienced significant delays and has not yet achieved effective broadband deployment for rural areas (Wireless Estimator, 2024). Such programs have led telecom providers to "cherry pick" more densely populated areas, thereby excluding many rural areas (Leighton & Clark, 2024).

Traditional satellite services, which in theory provide omnipresent internet connectivity, grapple with longer latencies and often slow or expensive data transfers (Hunter, 2021). Considering these factors suggest that one of the more effective approaches might be to implement a hybrid solution. In particular, the hybrid solution may consider the use of fiber optics as the network backbone while leveraging Fixed Wireless Access (FWA) technology to connect the final endpoints.

As the battle to capture the rural customer market intensifies, residents are presented with several wireless broadband options beyond satellite, namely mobile wireless and FWA. In any business, customer acquisition is usually one of the highest single-cost drivers, if not the most expensive, and many executives can detail their customer acquisition costs down to the dollar, if not the penny (Coleman, 2018). Accordingly, Internet Service Providers (ISPs) are zeroing in on both expanding their clientele and ensuring their loyalty, i.e., reducing churn. Churn is defined as the number of customers discontinuing their relationship with a company or service provider (Reichheld & Teal, 1996). In general, estimates for churn rate (i.e., the proportion of customers who defect during a time period) for local/regional ISPs often range from 1 to 5% per month (Wilson, 2021). Customers tend to leave when faced with reliability issues and lower speeds; however, improving service can be expensive (Ribeiro et al., 2023). Therefore, in the short term, it is more cost effective for FWA ISPs to strengthen their efforts to build customer loyalty. In the long term, effective churn management is a critical factor for the sustainability and success of a telecommunications company.

Fixed Wireless Access (FWA) providers face challenges in expanding their customer base due to the lack of incentives in current government initiatives such the BEAD. As a consequence, FWAs are less likely to attract new customers, making the retention of existing customers crucial to their operational success. Despite the lack of government incentives, FWAs will continue to play a significant role as one of the few available Internet Service Providers (ISPs) in many rural areas. Given this realization, some policymakers and industry experts have continued to advocate for the broader inclusion of FWAs in these government programs (Wireless Internet Service Providers Association, 2023).

The contributions of this paper are fivefold. First, this study improves our understanding of customer churn from models derived from primary and secondary data sources (i.e., identifying the determinants of churn). Second, it fills a gap in the extant literature; that is, no prior studies have comprehensively examined FWA churn. As such, when FWA studies were not available, studies that examined other ISPs were brought in to aid in model construction. Third, this study uses appropriate techniques that account for right-censorship (i.e., Random Survival Forests (RSF), KM plots, and Cox regression) to predict customers who are most likely to churn and ascertain variable importance. Fourth, the study provides a useful data template not only for older FWA ISPs but also those that have expanded into 5G. Fifth, the study emphasized the importance of maintaining healthy FWA providers, given their restrictions regarding BEAD subsidies.

The remainder of this paper is organized as follows: Section 2 provides a thematic literature review focused on the various lines of inquiry that informed our work. Section 3 introduces our conceptual model. In Section 4, we present the measures and exploratory analysis; this is followed by Section 5, which presents the descriptive analysis. In Section 6, we discuss the modeling methods. The results are reported in Section 7. Section 8 provides a discussion of the findings. In Section 9, we discuss the corporate and broader implications of our research. Finally, in Section 10, we discuss the study's limitations and offer our concluding remarks.

2. Literature review

2.1. The impact of churn

The enduring success of a business is tightly linked to its capacity for customer retention and its effectiveness in reducing customer turnover (Janzer, 2020; Mehta et al., 2016). Several studies have also shown that the cost of retaining past customers is about one-fifth of the cost of acquiring new customers (Athanassopoulos, 2000). Jain et al. (2021) highlighted that the costs associated with acquiring a new customer—advertising, promotional efforts, human resources, and time—can be considerable. With regard to telecommunication, many executives emphasized that customer retention is growing in importance (Pustokhina et al., 2021), and that addressing churn is becoming crucial for ensuring the survival of their organizations (Mattison, 2001).

2.2. Churn research

The field of telecommunications has numerous research papers addressing customer churn, as noted by Ahn et al. (2020). A significant portion of the extant research in this area is on the development of predictive models to determine which customers are more inclined to switch providers (Ribeiro et al., 2023). In a time when consumers are starting to have a wider selection of ISPs, the emphasis for these companies should be on bolstering loyalty and reducing customer churn.

Many factors can influence customer churn, such as service/product quality, economic and pricing considerations, competitive pressures, customer satisfaction and behavior, demographic elements, organizational policies, and others (Keaveney, 1995). For businesses, comprehensively tracking every potential item within these factors can be a daunting, if not infeasible, task. Consequently, a multitude of studies in this area are centered around churn prediction, and they prioritize feature selection based on the company's readily available data (Ribeiro et al., 2023). In contrast, past descriptive studies tend to spotlight variables most pertinent to the specific product or service being investigated. For instance, most descriptive telecom studies pinpointed price, age, gender, satisfaction, switching costs and barriers, and service quality as critical variables associated with churn (Ribeiro et al., 2023).

There is a dearth of churn research as it pertains directly to FWA ISPs. To overcome this gap, applicable churn research was drawn from three potential areas in order of preference. First, when possible, we used mobile internet studies, such as Gerpott and Thomas (2014). Second, when mobile internet studies would not suffice, we drew on general telecom studies, such as Ribeiro et al. (2023). Third, as a final option, we considered several omnibus studies, such as Keaveney (1995).

However, the applicability of studies to FWA from different industries and even within telecom can, in some instances, be questionable. For example, mobile telecommunications often have separate lines for plan members, and the mobile aspect may suggest caution for FWA comparison. Nonetheless, Gerpott and Thomas (2014) have noted that most mobile communications occur in fixed locations, suggesting some comparability to FWA. On the other hand, some mobile studies, such as Gerpott and Meinert (2018), were focused on usage and not specifically churn. That being said, an increase in usage is generally associated with a decrease in churn. This, in turn, implies the antecedents of usage as potential predictors of churn.

In this research, we will limit the review to the most relevant categories of variables, as dictated by the literature: the socio-economic variables of the household, the household's location, the local competitive market, product/service features, and, finally, customer behavior and satisfaction.

2.3. Household demographic determinants of churn

Household demographics refers to variables that characterize either the household as a whole or the individual members of the household. Additional household variables can include the number of members, the household's income, and the number of members responsible for the telecom bill. Individual demographics can be aggregated for the household as a whole, or they can be restricted to the person identified as the primary decision-maker.

Ribeiro et al.'s (2023) systematic review of the literature for telecommunications churn indicated that the household variables such as age, gender, and income had divergent results, while marital status and education were insignificant. In the following subsections, the review of socio-demographic variables is limited to these significant variables and a few other characteristics of the household.

2.3.1. Household demographic: Age

Age introduces a multifaceted dynamic when examining customer churn. While older customers, by virtue of their age, inherently have an extended window to exhibit loyalty, their slower adoption of the internet and potential reluctance toward newer technologies cannot be overlooked (Charness & Boot, 2009). Many studies (Becker et al., 2015; Garcia-Mariñoso & Suárez, 2019; Gerpott & Meinert, 2018; Jin, 2022; Lee, 2017) reported mixed results regarding the significance of age on churn, and other research has identified relationships between age and churn, but the directionality is inconsistent (Keramati & Ardabili, 2011; Lunn & Lyons, 2018; Seo et al., 2008; Su et al., 2012; Svendsen & Prebensen, 2013; Wong, 2011a).

In the context of mobile phones, a meta-analysis by Gerpott and Thomas (2014), discovered a weak correlation between age and usage. However, a more recent study by Kirgiz et al. (2024) showed that age is a strong predictor of churn behavior. Moreover, studies such as those of Chose and Han (2011) propose that the relationship between age and usage might not be linear; thus, correlation is not an appropriate measure.

2.3.2. Household demographic: Gender

Commonly held beliefs suggest that female consumers often exhibit a higher degree of loyalty, yet research indicates that customer loyalty between the genders is contingent upon the specific product or service in question (Melnyk et al., 2009). It may be hard to argue that ISP loyalty is influenced by gender, and, indeed, telecommunications research considering this aspect yields varied results. Kirgiz et al. (2024), Garcia-Mariñoso and Suárez (2019), Jin (2022), and Lee (2017) found no discernible gender differences in switching intentions or churn rates. However, research such as that of Becker et al. (2015) and Gerpott and Meinert (2018) did report disparities based on gender.

2.3.3. Household demographic: Income

The relationship between wealthier households and their behavior offers a varied landscape of findings. Broadly speaking, across service industries, affluent customers tend to be less affected by price increases (Keaveney, 1995; Kumar & Karande, 2000). As such, Reinartz and Kumar (2002) noted that such affluent customers demonstrate a diminished propensity to churn due to pricing. On the flip side for ISPs, Madden et al. (1999) found that those in lower income categories often display a greater likelihood of switching. Yet this latter observation is not universally consistent, as Lunn and Lyons (2018) found that income did not always have a pronounced effect on switching behaviors.

Gerpott and Thomas (2014) identified a direct link between higher income and increased usage. Increased usage could indicate increased loyalty; however, Chose and Han (2011) argued that the income/usage relationship might have complexities, suggesting it may not follow a simple linear trajectory.

2.3.4. Household demographics: Other

For telecommunications, minority households have historically lagged in their access to broadband internet. Specifically, a study by Prieger (2015) highlighted this disparity, illuminating the unequal access to broadband internet services. Interestingly, there seems to be a trend among minority groups where they tend to favor the use of mobile internet hotspots (Quintanilla, 2011). Given the economic challenges that some minority groups face, it is plausible that mobile hotspots present a more affordable and flexible alternative for them to stay connected instead of using FWA. However, the above is more indicative of adoption, not necessarily churn.

Additionally, larger household sizes have, surprisingly, exhibited a greater tendency to switch operators (Lee, 2017; Madden et al., 1999; Ribeiro et al., 2023). Within these households, the presence of children sometimes affects churn as well (Lunn & Lyons, 2018), but this observation is not consistently significant (Garcia-Mariñoso & Suárez, 2019).

Besides minority household status and the presence of children, other household demographics to consider could include: family structure (e.g., multi-generational households), lifestyle choices (e.g., preference for outdoor living), and preference for online shopping.

2.4. Household setting and competition

The locations of residences have a significant effect on telecom churn (Jin, 2022; Wong, 2011b). Location variables that indicate a higher concentration of people, such as apartment complexes and areas of denser population, can suggest an increased likelihood of a competitive offering (Conroy et al., 2021). Conversely, the higher likelihood of vacation homes, such as in rural Wisconsin areas, suggests a lower availability of internet services (Winkler et al., 2015).

Calvo-Porral et al. (2017) discovered that the appeal of competitive market options significantly influences customer churn within the mobile services market. Their analysis focused on the switching intentions of customers with and without contracts. It revealed that perceptions of attractive competing alternatives decreased retention in both cases.

Becker et al. (2015) analyzed competitor advertising spending—“TV, radio, press, online, direct marketing, billboards”—and its impact on churn for fixed internet services. They concluded that the level of customer exposure to competitive ads had a significant effect on churn. Keller (1993) discovered that a slight decrease in brand relationship has an impact on customer loyalty with respect to the principles of customer-based brand equity. More specifically, Svendsen and Prebensen (2013) indicated that positive brand image reduces churn.

2.5. Product/service characteristics

Product and service characteristics are among the most important reasons as to why customers churn (Alomari, 2017; Keaveney, 1995). These characteristics are obviously more specific to the situation at hand. For example, within telecommunications, Ribeiro et al. (2023) noted that churn issues related to “Service Quality, Satisfaction, and Service Value” had the greatest number of independent variables across studies. For these studies, the service quality construct’s impact on churn was observed indirectly through satisfaction.

Perception of service quality is one key determinant of customer satisfaction, where the latter can entail many other aspects, including need, emotions, expectations, and price (Zeithaml & Bitner, 2000). In general, customer satisfaction is well known to increase loyalty (see Dey et al., 2020; Garcia-Mariñoso & Suárez, 2019; Quoquab et al., 2018). Service quality issues tend to be more focused, including interactions at customer service touchpoints as well as core technology issues such as connectivity, reliability, and internet speed.

2.5.1. Price/discounts

Basic economic theory indicates that price materially impacts demand in competitive markets. Alomari (2017) found that price was by far the biggest reason for telecommunications churn and that service quality was a distant third. In fact, price discounts can be a common strategy to reduce the likelihood of churn for a specific customer (Ganesh et al., 2000), even though offering discounts alone is generally not as effective in restoring relationships as when they are combined with a service upgrade (Kumar et al., 2015). Consequently, generalizations about the importance of price need to proceed with caution. For example, Keaveney’s (1995) study, across many diverse industries, found that service was even more important than price.

2.5.2. Bundling/contracts

Offering service bundles can serve as a crucial strategy for telecommunication operators to reduce churn, as these bundles raise the switching costs for consumers (Üner et al., 2015). Switching costs act as a barrier, hindering customers from easily transitioning to alternative service providers (Su et al., 2012) and serve as a key element in fostering customer loyalty (Ahn et al., 2006; Kim & Yoon, 2004). Bundled services are now the primary method for accessing telecommunications services (Godinho de Matos et al., 2018). The better the quality of these bundled packages, the more satisfaction customers tend to have with their provider. This, in turn, fosters the development of trust in the company (Calvo-Porral et al., 2017).

2.5.3. Speed/reliability

In terms of network speed, Bhattacharyya et al.’s (2021) research on switching intention within mobile services revealed that an enhancement in network data speed leads to a reduced propensity for a customer to switch. It is reasonable to expect that this relationship, due to speed, also applies to FWA, as much mobile communication occurs at fixed locations (Gerpott & Thomas, 2014).

The reliability of telecom services is, understandably, an important factor in terms of increasing customer loyalty (Patrom, 2020). In fact, subscribers who chose their service provider due to network reliability showed a lower tendency to churn (Madden et al., 1999). However, continuing reliability issues are bound to be more problematic than with FWA given that mobile requires increased handoffs (or handovers), which can be a threat to reliability (Ekiz et al., 2005).

2.6. Customer behaviors and attitudes

Although one of the most predictive indicators of customer churn is the number of service complaints (Ahn et al., 2020), it is important to realize that these complaints are most likely due to failures in product and service characteristics, such as speed and reliability (Carnovale, 2024). Additional customer behaviors that may be important covariates of churn are the type and frequency of ISP usage (Madden et al., 1999). Needless to say, there are a great number of factors that can impact customer churn. Some antecedents not mentioned include customer tenure and unexpected charges (Lunn & Lyons, 2018). Given this extensive range of factors, it is not feasible to comprehensively assess all antecedents of FWA churn within the scope of a single study. Instead, it would be more practical to consult various literature review studies to gain a broader understanding of these influencing factors.

3. Conceptual framework

Based on the literature, a conceptual framework was developed in this paper to guide churn research within the context of FWA (see Fig. 1). This framework aims to present a holistic view for integrating disparate elements into a cohesive structure for data acquisition. Using the framework allows one to critically examine the data that is readily available from the company against variables that are typically used to predict churn. It should also be noted that acquisition of all elements in the framework represents an ideal situation, as it can be rarely achieved in practice (Ribeiro et al., 2023) .

The micro-level framework is comprised of six distinct categories of predictive measures: Household Demographics variables, Product and Service (P&S) characteristics, Customer Behavior/Attitudes, HH (Household) Setting, Competition, and Local Brand Presence. Within each category, one or more subcategories may exist; the subcategories are represented by ovals. Specific measures for each subcategory would be established at the time of modeling and would depend on the availability of data.

Within the HH groupings, both the Household Demographics and Household Setting are considered. To expedite the definition of these categories, examples of subcategories (variables) used to measure the parent groupings are given. HH subcategories include HH Size (e.g., number of people), Decision Structure of the FWA Account (e.g., joint account holder or otherwise), the HH Financial Status (e.g., total income), and Individual Demographics (e.g., age). The second HH category, Household Setting, considers two subcategories: Dwelling Location (e.g., population density) and Dwelling Type (e.g., single-unit dwelling or otherwise).

P&S characteristics are Price, Speed, Reliability, Contract/Bundling, and Other Service Attributes variables. The Price subcategory could include measures beyond plan rates, such as discounts, promotions, surcharges, etc. Internet speed (e.g., upload or download speeds) and service reliability (e.g., uptime) are core technology issues for ISPs. Two other subcategories include Contract/Bundling and Other Service Attributes (e.g., call-backs, wait time, etc.).

The Customer Behaviors and Attitudes category can include responses and interactions related to the P&S issues mentioned above. The Other Behaviors subcategory pertains to internet usage. Considerations within this subcategory include frequency of usage and type of usage (e.g., online shopping, video streaming, etc.). Lastly, customer attitudes have been shown to be strongly related to churn, but the availability of such measures for all customers is doubtful. For example, customer satisfaction studies are usually done for only a subset of customers. Other attitudes may only be recorded at the survey time, at the time of an incident, or upon service termination.

The competitive category includes alternatives such as other wireless providers and landline services. Landline offerings can include DSL, cable, and fiber. However, cable and fiber options are obviously less present in rural settings. Wireless competition includes other FWA providers, satellite, and cellular data services. Unlike cable and fiber, satellite, and, to some degree, cellular data services, are more or less omnipresent. Inclusion of omnipresent variables (i.e., no variation) is not informative to a model. Alternatively, these variables could be represented using a measure with some variability, such as maximum speed in that locality for that service.

The last category is Local Brand Presence. Many companies are divided into sales territories where the regional managers may have some autonomy in running operations. Consequently, the subcategory Sales Zone Characteristics could include many variables specific to the company, such as types and levels of advertising, promotions, and other services.

4. Measures and exploratory analysis

4.1. Regional cell tower company

This study sought to better understand what the determinants of customer churn were for a company offering FWA. To explore this, Bug Tussel (BT), a regional ISP headquartered in Green Bay, Wisconsin, was chosen. BT was established in 2003, and its primary mission has been to offer dependable wireless internet connectivity, predominantly serving rural customers. BT has about 600 tower

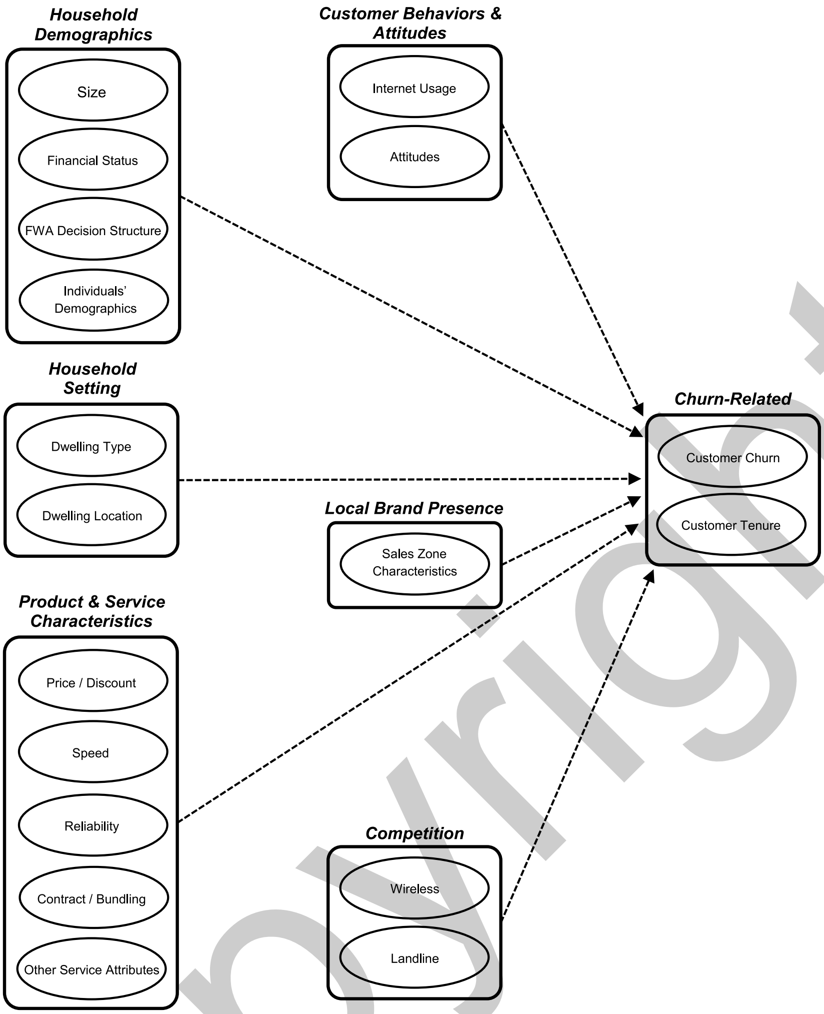


Fig. 1. Micro-level conceptual data framework of FWA churn.

locations; most are within Wisconsin, but it also has a presence in Northern Michigan and Iowa. The towers serve a dual purpose, as they are used for both FWA and cellular communication services.

The customers included in this study had an active ISP account at some point between January 1, 2018, and March 1, 2023.[1] This timeframe was chosen for a number of reasons. First, it is a reasonable length of time to be able to conduct a churn analysis. Second, the start of this timeframe marks a period where the company's information systems were improved to the point where external data sources could be used to supplement the company's data. Note, there was a small percentage of customers where external data was not available; thus, those records were omitted. Third, the technology employed on the towers could be regarded as relatively recent throughout this timeframe.

For BT, it was vital to comprehend how various factors—product and service features, socio-economic elements of the household, local brand presence, and competition—impacted customer retention. This knowledge would not only be key for reducing churn, but it would also be valuable for crafting marketing strategies in a dynamic, competitive environment.

4.2. Voluntary versus involuntary churn

For churn studies, it is normally the case that the customers associated with involuntary churn are removed from the dataset, leaving 7334 records. The vast majority of those involuntary churn cases involved customers moving out of the service area and financial difficulties. Other causes of involuntary churn included death, incarceration, or house fires. In contrast, the top reason for

[1] This timeframe includes the COVID-19 pandemic era, which undoubtedly impacted demand (Pflughoeft et al., 2022); however, our model focusses on microlevel variables, and the exploratory analysis did not indicate a significant change regarding churn.

voluntary churn was defecting to a competitor. Other circumstances regarded as voluntary churn included failing to provide payment for service without formally canceling (BT does not have service contracts), and no longer needing internet service.

4.3. Model variables

Using the framework described in Section 3, this study will use the relevant subcategories where BT can obtain data both internally and externally. The response variables and the available predictors used in the model are highlighted in Table 1. The table shows the variables that were used to measure each of the relevant subcategories within the framework. The parent categories associated with each subcategory, the data sources used, and the hypothesized impact on churn are also shown in the table. The second-to-last column identifies internal (labeled as BT) and external data sources (third-party data provider), the U.S. Census, and the FCC. The last column was used to show the hypothesized impact on churn. A footnote of “c” denotes a direction proposed by the authors of this study, while “a” indicates a directionality claim derived directly from the literature, and “b” signifies an inference based on the literature. An example of a variable that is drawn directly from the literature is number of adults in the household (Lee, 2017; Madden et al., 1999). An example of a variable whose impact is inferred would be credit card, as having a credit card is in some ways a proxy for the financial stability of the household, as, for the most part, the literature suggests that a more financially stable household would be less likely to churn (Reinartz & Kumar, 2002).

Out of the six categories of measures, only one was excluded: Customer Behaviors/Attitudes. As with most companies, customer attitudes for BT are surveyed only periodically for a subset of customers, thus limiting its value for modeling churn. It should also be noted that BT’s Customer Behavior data, including usage data, was lacking due to deficiencies in their Customer Relationship Management (CRM) system as well as some technical and privacy issues.

Additionally, there were a few subcategories (i.e., ovals) within the Product and Service Characteristics category that were not modeled: Contract/Bundling, Price, and Other Service Attributes. Contract and Bundling are not applicable to BT’s ISP offering. With price, there is almost a one-to-one correspondence with speed; thus, only one of those variables could be used. The Other Service Attributes subcategory is meant to capture considerations that are specific to an organization and could be modeled when applicable.

Other notable BT-sourced variables were Price Discount, Sales Zone, and Distance to BT Tower. The Price Discount variable was used to capture some of the explanatory power related to pricing (Note: transformations of price and discount, such as net price and discount percentage, led to problems in the survival models and thus were no longer considered.) The Sales Zone variable originally contained 13 zones, but they were collapsed into five categories[2] for predictive modeling. The Distance to BT tower variable represents the closest tower to the customer, which is a rough proxy for reliability. The rationale for this lies in the fact that the further the user is from the base station, the greater the attenuation of the signal, reducing the reliability of communication services (Venkatesulu et al., 2016).

The FCC data sources offered insights into ISP competition. First, they provide tower location, enabling the calculation of the distance to the closest competitor. Second, FCC Application Programming Interfaces (APIs) can identify the U.S. Census Block ID for a household, which helps determine the presence of competitive alternatives (i.e., DSL, Cable, and Fiber) in the immediate region. The FCC data has faced criticism in the past for assuming universal access within a census block, even if only one household has access to it. Nonetheless, it is still a useful variable, and the only one available one from a historical perspective.[3]

The data acquired for the third-party data provider included much information about the Household Occupants.[4] This data was attained in a secure manner and used in a manner that falls under “fair use.” The variables provided were Number of Household Adults, Presence of Credit Cards (determined from all types), and Age of Primary Decision-Maker. The data from the third-party data provider also contained information on single versus multi-unit homes, which is an attribute of the Dwelling Type subcategory under Household Setting.

In addition to Dwelling Type, the Household Setting category includes data sourced from the U.S. Census, which allowed for the calculation of population densities. Denser population can also suggest an increase of competitive services, such as cable and fiber. Population density is calculated at the census tract level using land area only (i.e., not including bodies of water). Specific information on all model variables is presented in the following section.

5. Data considerations

5.1. Model variables

Statistics for all model variables and price, using voluntary churn records, are presented in Table 2. For numeric variables, the range and the mean are provided, while for indicator variables, the percentage of observations that were coded “Yes” are presented.

With respect to numeric variables, our typical customer profile reveals an average Tenure of 1280 days (about 3.5 years). Predominantly they are older individuals (53 years), usually living in households comprised of two members, with household incomes of about $73,000. They typically reside in rural areas that have about 25 customers per square mile. The average plan price was $58.46 per month. The longest Distance to a BT tower was 13.97 miles; however, 98% of customers were within 5 miles of a BT tower.

[2] The zone groupings are adjusted to a particular modeling method.
[3] For more recent data, the FCC has shifted to Uber’s hierarchical hexagonal methodology (Brodsky, 2018).
[4] The third-party data provider data did not include levels of education or race of the occupants.

Table 1
Variables within the framework for FWA churn.

Category	Subcategory	Variable (Increase/Presence of) Leading to the Hypothesized Impact on Churn	Source	Hypothesized Impact on Churn
Household Occupants	Size	Number of Adults	Third-party dataprovider	Increase[a]
	Financial Status	Income	Third-party dataprovider	Decrease[a]
		Credit Card (present)	Third-party dataprovider	Decrease[b]
	Decision-Making	Joint Account (present)	BT	Decrease[c]
	Individual Demographics	Age	Third-party dataprovider	Decrease[a]
Household Setting	Dwelling Type	Multi-Unit (present)	Third-party dataprovider	Increase[b]
	Dwelling Location	Population Density	U.S. Census	Increase[a]
Product and Service Characteristics	Internet Speed	Speed	BT	Decrease[a]
	Reliability	Distance to BT Tower	BT	Increase[b]
	Price Discount	Discount	BT	Decrease[a]
Local Brand Presence	Sales Zone Characteristics	Sales Zone	BT	Used for Stratification
Competition	Wireless	Competitive Distance to Tower	FCC	Decrease[b]
	Landline	DSL (present)	FCC	Insignificant[b]
	Landline	Cable (present)	FCC	Increase[b]
	Landline	Fiber (present)	FCC	Increase[b]
Churn-Related	Customer Churn	Churn	BT	N/A[d]
	Customer Tenure	Tenure	BT	N/A[d]

Note: [a] directly from the literature; [b] inferred from the literature; [c] proposed by authors of this study; [d] response variables.

Table 2
Statistics for numeric and indicator variables.

Type	**Variable**	**Range**	**Mean**	**Model**
Numeric	Tenure (Days)	1–5308	1280	Yes
	Age (Years)	18–99	53.05	Yes
	Number of Adults	1–6	1.99	Yes
	Income ($)	12,500–142,500	73,320	Yes
	Population Density (per sq. mile)	1.4–1642	25.03	Yes
	Distance to BT Tower (miles)	0.02–13.97	1.99	Yes
	Distance to Comp. Tower (miles)	0.02–5.72	1.51	Yes
	Speed (Mbps)	10–100	17.76	Yes
	Discount ($)	0–75	7.76	Yes
	Price ($)	45–190	58.46	No
Type	**Variable**	**Percent**		**Model**
Categorical Indicator (Binary)	Sales Zone (1, 2, 3, 4, 6, 7)	96.1%		Yes
	Credit Card	92.6%		Yes
	Joint Account	30.7%		Yes
	Single-Unit	97.4%		No
	DSL	37.7%		Yes
	Cable	14.1%		Yes
	Fiber	6.9%		Yes
	Churn Rate [a]	27.4%		Yes

Note: [a] for the entire period.

Similarly, all of the competitive tower distances were within 5.72 miles of the residence.

For indicator variables, we noted the following: most customers (97%) live in a single-dwelling home; the vast majority, 92.6%, have a credit card; and 30.7% have joint accounts. Within their census block, the competing landline options of DSL, Cable, and Fiber were available at the rates of 38%, 14%, and 7%, respectively. In total, 43.5% of customers were in Sales Zones 1 & 2, which are BT's oldest and most established zones.

5.2. Right-censored data

Since this is a temporal study with regard to churn, it is important to handle right-censored data in a reasonable manner. Right-censored data refers to situations where the event of interest (i.e., churn) has not yet been observed for all customers. In other words, all customers will eventually churn; however, since the study must have a predefined analysis window, churn will not be observed for all customers. Failure in addressing right censorship within long-term studies may introduce significant biases (Turkson et al., 2021), such as being overly optimistic about customer retention rates and incorrect estimation of feature importance. Using survival methods that are designed for these situations provides a better holistic picture than other techniques, such as logistic regression and Random Forest (RF), which are acceptable techniques for short-term predictions. In other words, survival techniques predict not only if a customer will churn, but also when a customer will churn. Although survival analyses are often employed in the medical field, these methods are still used in other areas (Allison, 2014; Moore, 2016).

6. Modeling methods

For our study, we had two distinct objectives: to create an overall model with strong predictive capabilities and a separate explanatory model to provide further insights. With regard to the first objective, our aim was to create a predictive model that would not only determine the customers most at risk for churning at specific points in the future, but also identify the most important predictive variables. To achieve this, we used Random Survival Forests (RSF), a machine-learning technique within survival analysis (Ishwaran et al., 2008). An RSF is a nonparametric, flexible ensemble technique that makes fewer structural assumptions as compared to traditional survival methods. RSFs are similar to a random forest method, including their ability to estimate variable importance. Another extension of RSFs is that they can estimate a cumulative hazard function, which is essential for analyzing time-to-event data (Rezaei et al., 2020).

An RSF is a data-driven approach that relies on a collection of decision trees. Many decision trees can be created for a single sample, as the technique relies on bootstrapping. On average, each bootstrap selects approximately two-thirds of the original records, whereas the unselected records can be used to estimate out-of-bag prediction error, lessening the need for a training and a testing dataset. By utilizing a set of bootstrapped decision trees, RSFs are able to handle higher-dimensional data and their interactions, as well as provide importance estimations. More information regarding random forests can be found in the work of Breiman (2001), while Ishwaran et al. (2008), Wang and Li (2017), and Rezaei et al. (2020) provide discussions specifically on the RSF technique. Moreover, Kirgiz et al. (2024) provide an example of random forest application in the context of telecom and Routh et al. (2021) demonstrate an application of RSF for churn.

To evaluate the performance of the RSF, the concordance index (i.e., C-index) can be calculated. C-index is a common measure of model fit (for further description see; Wen et al., 2022). The C-index is an extension of the area under the ROC curve (Heagerty et al., 2000). The C-index ranges from 0 to 1, with higher values indicating better fit; for binary outcomes, C-index and Area Under the Curve (AUC) values can be numerically equivalent or very close. Since RSFs are a time-to-event analysis, a time-varying AUC can be created to estimate the risks of different times in the future. Additionally, the AUC value of a logistic regression at Time 0 is calculated as a baseline for comparison. The logistic model would be used only for prediction, and it has the same set of independent/dependent variables with the exception that customer tenure is treated as an independent variable.

The second objective of this study was to develop a more insightful explanation beyond variable importance by exploring the Kaplan-Meier (KM) Survival Plots and the Cox regression model. KM plots are used primarily in survival analysis to graph the survival function for the individual covariates. This plot helps visualize the probability of an event (such as churn) for each predictor over time. Furthermore, KM plots provide the ability to examine different groups within predictors to determine if they have different survival probabilities (e.g., low- versus high-income groups). Formally the differences between the “designated” groups can be tested using a

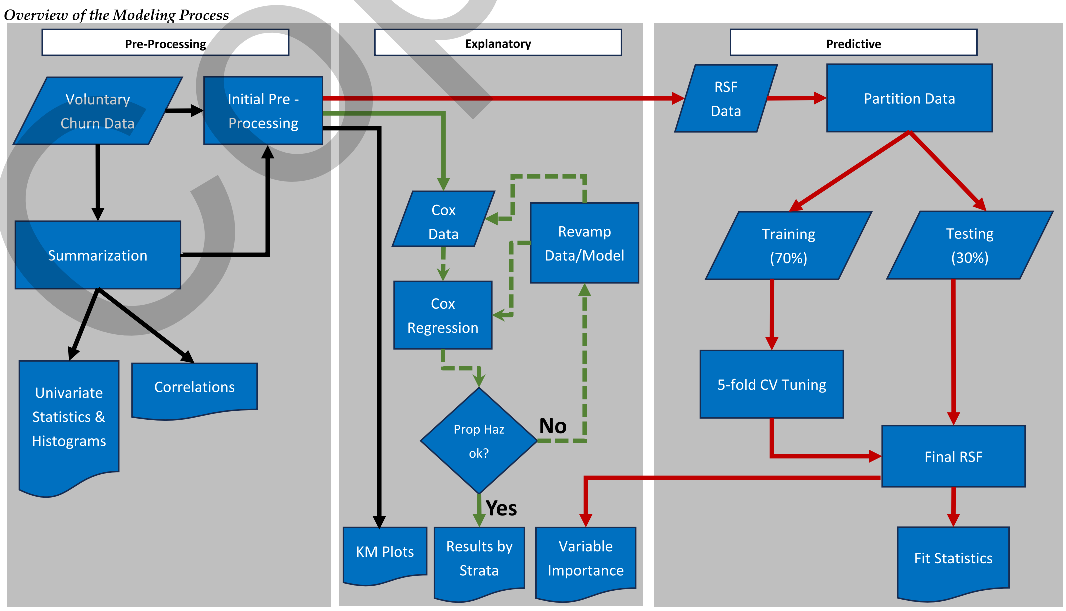


Fig. 2. Overview of the modeling process.

standard two-tailed Log Rank Test. However, it is important to note that the results of the KM log-rank test, may not always be consistent with the results of a Cox regression due to various factors. For example, a KM log-rank test requires a somewhat arbitrary grouping within a specific covariate, and it does not consider the impact of other covariates. Nonetheless, a strong effect of the covariate on churn would most likely be evident by the test.

A Cox model is used to perform survival analysis in a multivariate fashion, which is considered the gold standard for such analysis (Bice et al., 2020; Diana et al., 2019). Unlike a logistic regression, which is only concerned with the absolute probabilities at a fixed point, a Cox regression focuses on the relative risk over time. Consequently, a Cox regression gives a more detailed picture of the processes leading up to the event, not just whether the event happens or not. Using a Cox model offers several advantages. First, it does not require a specified distribution for survival times. Second, it provides a highly interpretable framework for the results in the form of a regression between survival time and explanatory variables (Schober & Vetter, 2018). The Cox regression model is grounded in the hazard function. Mathematically the Cox model is expressed as follows:

$$H(t) = H_0(t) \times \exp(b_1x_1 + b_2x_2 + \cdots ..b_kx_k).$$

The Cox model does have some stringent assumptions concerning proportional hazards where violations can potentially bias the results. Consequently, it becomes difficult to develop high-dimensional models as each covariate, as well as the model, must pass the proportionality assumptions. These assumptions can be tested by the use of Schoenfeld residuals. As such, model development becomes an iterative process to determine what set of predictors can be retained. Once the final model is developed, the exponentiated beta coefficient allows the researcher to assess the relative changes in the hazard corresponding with a specific unit increase in the covariate. The fit of the model will be assessed using only a C-index value, as the model is only for explanatory purposes. This model is not used for predictive purposes and thus uses all the data just like the KM plots.

The modeling process is briefly summarized in Fig. 2. As noted in the figure, this process was composed of three stages represented in gray rectangles: pre-processing, explanatory, and predictive. In the initial stage, pre-processing was done largely to assist in the modeling efforts for the voluntary churn data. The data were inspected for anomalies; few were found and were subsequently corrected. Correlation among predictors was checked for multicollinearity, which could adversely impact the explanatory model or the variable importances derived from the predictor model. Additionally, univariate statistics and histograms were generated for a variety of purposes. For example, a categorical variable with numerous levels may lead to artificially high RSF importance, or variables with little variation would lead to minimal importance. In addition, histograms/tables were helpful for identifying categorical variables with observations per level, which would be needed to be combined for analysis.

The predictive stage is used for the development of the RSF, which can identify nonlinear relationships. The RSF modeling results will hinge on a single 70/30 split for training and testing purposes; however, the tuning of the model uses five-fold cross validation. The final model selected is the one whose hyperparameters led to the highest average Out-of-Bag Concordance Index (i.e., C-index). Three major hyperparameters were tuned in this study, namely the number of variables tried at each node, the number of trees, and the minimum number of observations required to split a node. This tuning was done via *mtry*, *nodesize*, and *ntree* within the R package *randomForestSRC*. Dauda (2022) identified these three to be the major, central hyperparameters to be tuned.

The explanatory stage is driven by the results from the multivariate Cox regression. Besides the Cox regression, the explanatory stage utilizes the RSF variable importances as well as the KM Plots. The latter is especially useful when variables must be excluded from the Cox model due to non-proportionality. Collectively the aim of these stages was to provide a holistic approach to analysis.

Invariably the entire modeling process (i.e., the Cox regression and RSF models) is more complicated than displayed in Fig. 2. To this point, there is some interplay that can, and does, occur between the stages. However, this figure does communicate the essence of the process.

Table 3
Overview of the multi-method results.

Variable (Increase or presence of)	Hypothesized Impact on Churn	KM Log Rank Test (All Zones)	Cox Regression (Zones 1-4, 6, and 7)	Consistent with Hypothesis	RSF Importance (All Zones)
Fiber (present)	Increase	Sig, Inc	Sig, Inc	Yes	18.02%
Sales Zone	Significant	Sig	Stratified	Yes	17.23%
Age	Decrease	Sig, Dec	Sig, Dec	Yes	16.65%
Discount	Decrease	Sig, Dec	N/A	Yes	11.79%
Speed	Decrease	Sig, Dec	Sig, Dec	Yes	7.71%
Credit Card (present)	Decrease	Sig, Dec	N/A	Yes	4.70%
Pop. Density	Increase	Sig, Inc	N/A	Yes	4.40%
Nbr. of Adults	Increase	Sig, Dec	N/A	No	4.21%
Income	Decrease	Sig, Dec	N/A	Yes	3.72%
Multi-Unit (present)	Increase	Sig, Inc	N/A	Yes	3.44%
Dist. to BT Tower	Increase	Insignificant	Insignificant	No	2.41%
Dist. to Comp. Tower	Decrease	Insignificant	Sig, Dec	No	2.31%
Cable (present)	Increase	Sig, Inc	N/A	Yes	1.60%
DSL (present)	Insignificant	Sig, Inc	N/A	No	0.93%
Joint Acct. (present)	Decrease	Sig, Dec	Sig, Dec	Yes	0.89%

Note: consistency checks were made among the hypothesized relationships (Column 2), the results of the KM Log Rank Test (Column 3), and the Cox regression (Column 4). The outcome of the consistency checks is reported in Column 5.

7. Results

7.1. Overview of the multi-method results

Table 3 provides a summary of the multi-method results as compared to the hypothesized relationships. For example, the presence of Fiber was hypothesized to have an increase on churn. Both the KM log-rank test and the Cox regression indicate that this variable in fact increases churn, which is consistent with the hypothesis. The last column of RSF importance shows that fiber is the most important variable in predicting churn, at 18%.

In general, the variables identified with a “Yes” in Column 5 indicate consistency. Findings were not considered conflicting when there was an N/A in the Cox regression column. Due to proportionality issues, only six variables could be modeled in the Cox regression: Fiber, Age, Speed, Joint Account, Distance to BT Tower, and Distance to Competitive Tower. The first four variables for the Cox regression mentioned above were consistent with the hypothesized impact on churn. The Distance to Competitive Tower was significant in the Cox regression, but insignificant in the KM plot. Distance to BT Tower was insignificant in both the KM plot and the Cox regression. The consistent variables–Fiber, Age, Speed, and Joint Account–accounted for about 43% of the RSF importance. For the RSF, the top four variables for predictive importance are: Fiber, Sales Zone, Age, and Discount.

The variables that were only in the KM plots (i.e., not in the Cox regression) were Sales Zone, Discount, Credit Card, Population Density, Number of Adults, Income, Multi-Unit, Cable, and DSL. All except Number of Adults and DSL were consistent with the hypothesized impact on churn. An increase in the Number of Adults, as indicated in the literature, was hypothesized to increase churn, but our results showed otherwise. DSL was anticipated to not have any impact on churn, but the KM plot indicated that its presence significantly increased churn. These two latter variables only represented 5.14% importance in total. In the following subsections, we provide a more detailed analysis of the results for each of the methodologies.

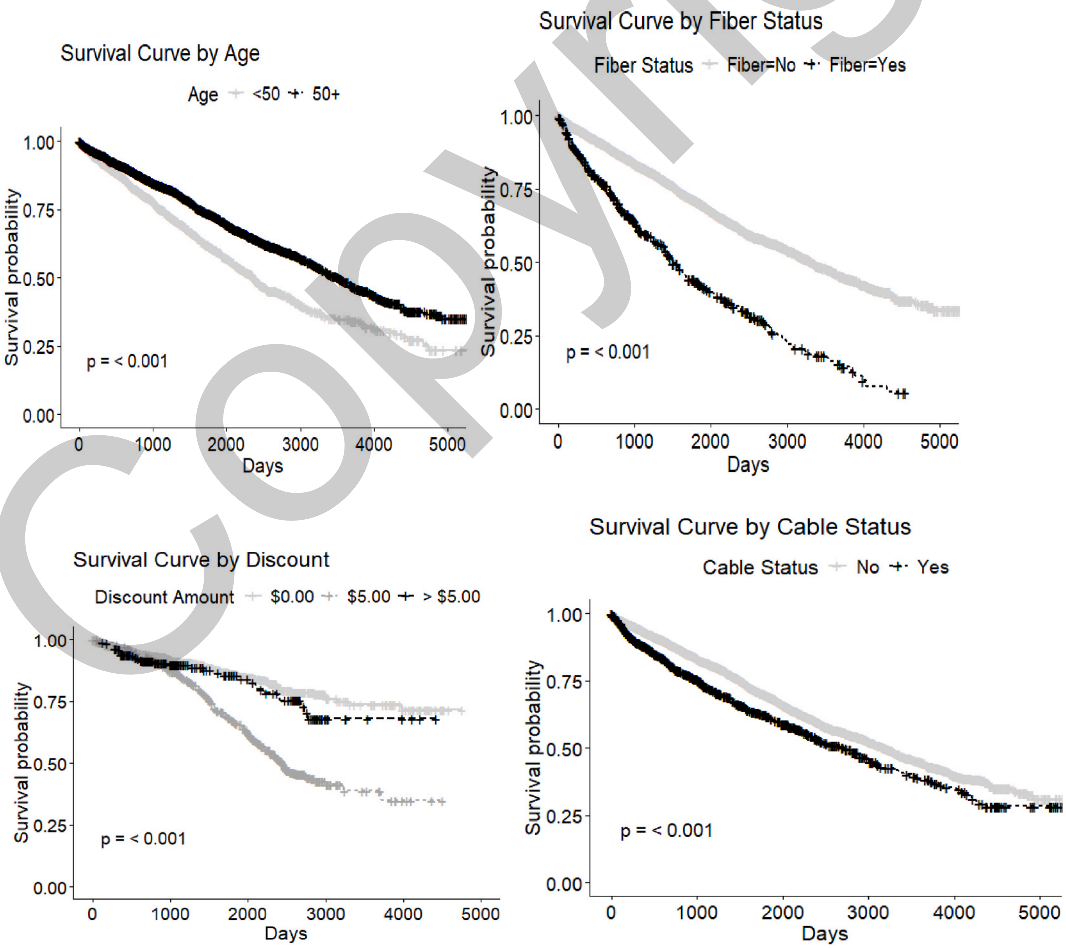


Fig. 3. KM Plots, with log rank tests, for Age, Fiber, Discount, and Cable.

7.2. KM plots

KM Plots, with Log Rank Tests, were examined for all 15 predictors. These plots are useful for understanding the impact of an individual variable against churn and survival time. Fig. 3 shows the plots for four predictors: Fiber, Age, Cable, and Discount. It was evident from the distance between the survival curves in the Fiber KM plot that customers with fiber service experienced notably different retention outcomes than those without. In other words, the presence of Fiber had a profound impact on churn, as shown by the wide separation of the survival probabilities over time and significant log rank test. In contrast, the separation of the survival curves for Age and Cable plots were more subtle, suggesting a smaller effect on churn. In terms of Discount, it is evident from the KM plot that there is a meaningful difference between no discount and some discount, but not so much with regard to the amount of discount. The KM plots for all the other variables in this study can be found in the appendix.

7.3. Random survival forest model

For the RSF, all records were used for analysis, although 70% of the records were used for tuning and 30% of the records were for testing purposes. The final RSF model used for prediction and importance consisted of 1000 trees (*ntree*), whereas the *mtry* and *nodesize* hyperparameters were 6 and 15 respectively. The C-index for the testing data was 0.719, which indicates that the model has a reasonable level of fit. In general, a C-index of about 0.7 is considered good (DeMaris & Selman, 2013; Jin et al., 2020; Steyerberg, 2009) and indicates that the model is reasonably able to differentiate between individuals with different survival times.

In addition to the C-index measure, the average AUC value can be calculated for various time points in the future, as shown in Fig. 4. For time 0, the AUC measure was 0.72 and did not vary much. For 360 days out, the AUC value rose to 0.73. The dotted lines represent a 95% confidence band, and the small line segment on the left side of the graph represents the AUC (0.76) from a comparable logistic regression model.

The variable importances (Ishwaran , Lu, & Kogalur, 2023) for the RSF model are shown in Fig. 5, which were transformed to sum to 100%. The three most important variables were Fiber, Zone, and Age, with values of 18.02%, 17.23%, and 16.65%, respectively. The five least important variables were Distance to BT Tower, Distance to Competitive Tower, Cable, DSL, and Joint Account, with values of 2.41%, 2.31%, 1.60%, 0.93%, and 0.89% respectively. Since variable importances can be impacted by multicollinearity, the top seven inter-predictor correlations are shown in Table 4. Most of these correlations are relatively weak, except for Discount and Speed, as well as the distances between BT Towers and Competitive Towers. From the importance plot, we can see that these two pairs of importances are adjacent to one another in their respective pairs. The tower distances are relatively weak importances, but discount and speed are a bit more substantive.

The Discount & Speed correlation will cause some ambiguity in their RSF importances, but maintaining some price component within the model was deemed necessary. The correlation between the Tower Distances is moderate, but that, and the remaining inter-predictor correlations, were not high enough that the importance values should be substantively impacted or diluted (i.e., all predictors were included for the RSF).

7.4. Cox regression model

Extensive analysis showed that it was not possible to represent all the sales zones into one Cox regression model while still strictly adhering to the proportionality assumptions. Consequently, the data were stratified by zones to create separate Cox regressions. In the interest of brevity, the detailed results of only one Cox regression are reported in this paper. Sales Zones 1, 2, 3, 4, 6, and 7 were used, resulting in a final sample size of 7056–which represents about 96% of the observations, with a churn rate of 27.4%. Not all of the 15 predictors could be used in the Cox model. Nine variables needed to be excluded due to reasons such as low frequences, violations in the proportionality assumptions, stratification (e.g., zones), or multicollinearity (e.g., Discount and Speed). The Cox regression results

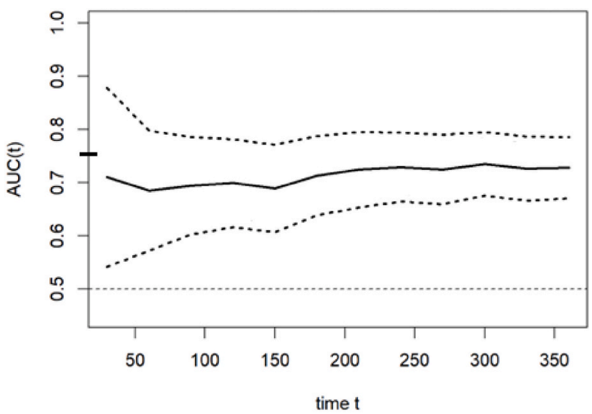


Fig. 4. Time-dependent AUC (RSF).

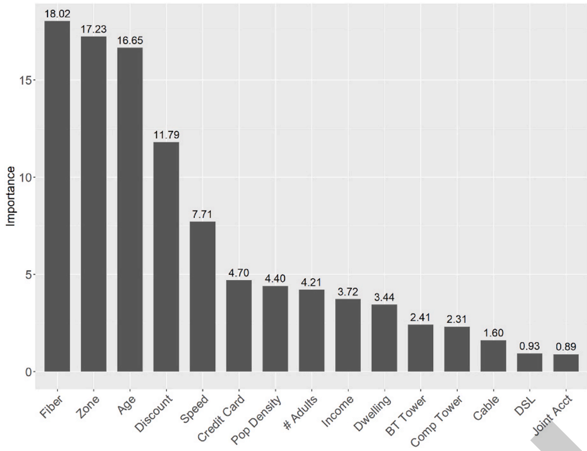


Fig. 5. Rsf model importances.

Table 4
Top seven inter-predictor correlations based on absolute values.

Pairwise Relationship	Correlation
Discount & Speed	0.830
Distance to BT Tower & Distance to Competitor Tower	0.530
Number of Adults & Credit Card	0.280
Cable & Pop. Density	0.223
Cable & Distance to Competitor Tower	–0.163
Joint Account & Number of Adults	0.178
Distance to BT Tower & Pop Density	0.159

are shown in Table 5, which lists the covariates, their coefficients, the exponentiated coefficients, the Z-tests, and their *p* values. All covariates, except for BT Tower Distance, were statistically significant ($p < 0.05$). For the remaining covariates, exponentiated coefficients less than one indicates a lesser hazard ratio. All variables except BT Tower Distance were significant, for an alpha of 0.05. It should be noted that the beta coefficients in a Cox regression are relatively stable with slight variations in the inclusion or exclusion of variables (Babalola & Yahya, 2019).

Chi-square tests were done on the Schoenfeld residuals for all covariates and are shown in Table 6. None of the associated *p* values was less than 0.05, indicating that the proportionality assumption could not be rejected for any covariate as well as for the model (i.e., GLOBAL).

8. Discussion

In this section, we discuss the findings of our research and make connections to our hypothesis and to some previous studies in this area. We will also offer interpretations of our findings, providing insight into why certain results may have emerged. The findings will be discussed in the context of the three approaches: KM Plots, Cox regression, and RSF variable importance. Although there is considerable consistency among the results, the reader should keep in mind that the above approaches will yield slightly different results; this is due to the different methodologies and/or the different subsets of data for these methods. The KM Plots and the Cox regression provide the clearest explanatory results, and the RSF can be used for variable importance. Because the Cox regression model provides a multivariate view, a brief summary of the results is presented in Table 7 based on the exponentiated beta coefficients. The results will be discussed in relation to the subcategories (i.e., ovals) that appear in our conceptual framework (see Fig. 1). Additionally, we will discuss potential future directions for research based on our findings and highlight any limitations that may have influenced

Table 5
Cox regression results for zones 1, 2, 3, 4, 6, and 7.

Predictor	Coef	exp(coef)	Z test	p value
Joint Acct (No)	−0.3407508	0.7112361	−6.878	<0.001
Age	−0.0214167	0.978811	−12.49	<0.001
Speed	−0.5186597	0.5953179	−4.903	<0.001
BT Tower Dist.	0.0005648	1.0005649	0.026	0.9795
Dist. to Comp. Tower	−0.0680404	0.9342227	−2.166	0.0303
Fiber	0.8307226	2.2949766	12.843	<0.001

Table 6
Chi-square test results for scaled schoenfeld residuals.

Covariates	chi-sq	df	p value
Joint Acct	3.058	1	0.08
Age	2.719	1	0.10
Speed	0.164	1	0.69
BT Tower Dist.	1.615	1	0.20
Dist. to Comp. Tower	3.444	1	0.06
Fiber	0.207	1	0.65
GLOBAL	9.245	6	0.16

Table 7
Cox regression results for zones 1, 2, 3, 4, 6, and 7.

Variable	Interpretation[a]
Joint Acct	Customers who have a joint account have a 28.9% decrease in churn risk
Age	Customers who are 10 years older have a 19.3% decrease in churn risk
Speed	For customers who go from low speed (<50 Mbps) to high speed, there is a 40.5% decrease in churn risk
BT Tower Dist.	Insignificant
Dist. to Comp. Tower	For each 1-mile increase, there is a 6.6% decrease in churn risk
Fiber	The presence of fiber increases churn risk by 129.4%

Note: [a] as measured by the hazard ratio.

our results.

8.1. *Characteristics of Household Occupants*

Within the context of mobile communications, the present analysis of customer churn yielded several intriguing insights regarding the impact of household factors (i.e., Age, Credit Card, Income, Number of Adults, and Joint Status). From the KM Plots, all of the household occupant variables showed significant differences in isolation. The KM Plots also revealed that customers who were older, possess a credit card, have a joint account status, reside in households with multiple adults, and earn higher income tend to exhibit greater loyalty. When looking at RSF variable importances, age is the third most impactful, at 16.65% of total importance, while each of the remaining household occupant variables has less than 5% importance.

According to the multivariate results of the Cox regression, only two of the household occupant variables, namely Age and Joint Account, satisfied the proportionality assumption. As shown in Table 7, the Age variable was significant, showing that a 10-year increase in age is associated with a 19.3% decrease in churn risk. Joint Accounts were also significant and led to a 28.9% decrease in churn.

As with previous studies, our findings suggest there is a relationship between age and churn (Garcia-Mariñoso & Suárez, 2019; Gerpott & Thomas, 2014; Jin, 2022; Lee, 2017). More specifically, Lunn and Lyons (2018) found that those who are retired are less likely to churn and are less aware of the speed of their internet provider (Newswire, 2021). Other studies have also shown that older adults may prioritize service stability and are less likely to switch providers due to unfamiliarity with newer technologies or the perceived effort associated with changing services (An et al., 2024).

Moreover, in the U.S., 17.5% of individuals living in rural areas were aged 65 and older, compared to 13.8% of those residing in urban areas (Smith & Trevelyan, 2019). This percentage is even more pronounced in Wisconsin, where 31% of the rural population is age 65 years or older. It is logical to assume that our finding concerning loyalty among older customers is linked in part to their lack of awareness of issues concerning rural internet service providers.

For the other remaining household occupant variables that were excluded from the Cox regression, the results from the KM Plots and RSF variable importance were used for interpretation. With respect to the number of adults in the household, the KM Plots were not consistent with the previous literature. Studies such as Lee (2017) and Madden et al. (1999) suggested that households with multiple family members tend to switch more frequently, while our results showed just the opposite. The RSF results suggest that this variable is of limited importance in a predictive sense (i.e., <5%).

As for proxies of household wealth—namely household income and credit card accounts—the KM Plots indicated that more affluent customers are more loyal. Surprisingly, the impact of household income on churn has not been clearly established in past telecom research (Burnett, 2014; Lunn & Lyons, 2018). However, the broader consumer behavior literature highlights that higher-income households may have a greater capacity to pay for premium services, and they are less influenced by price changes (Kamakura & Mazzon, 2015). Additionally, higher-income customers often seek stability and convenience, which can act as deterrents to churn (Kotler & Keller, 2016).

Credit cards can be a weak proxy for household wealth, and they can be tied to automatic payment systems, which create barriers to churn (Huang et al., 2015). For BT, the presence of a household credit card led to a significantly less churn rate despite the fact that BT did not have an automatic payment system for many years. Even so, it is logical that the use of a credit card for payment, done over the phone, led to less churn, as it may be more convenient than the use of a personal check. For our data, the predictive importance of a credit card is less than 5% RSF importance, most likely reflecting the situation that only 7.4% of BT households do not have a credit card.

8.2. Household setting

Variables related to the household setting, such as the type of dwelling and population density, were significant in the KM Plots, contributing 3.4% and 4.4% RSF importance, respectively. Despite multi-unit dwellings being more appealing to ISPs due to their higher potential customer base, the KM Plot revealed that single-unit dwellings were much more loyal. This is most likely due to the stability of the residence in single-unit dwellings (Rohe & Stewart, 1996). It is also important to keep in mind that only 2.6% of dwellings were multi-units.

In the KM Plot, population density was grouped as ≤25 residents per square mile versus otherwise. The survival curve for the lower-density areas had a lower risk of churn. This stands to reason, as higher-density areas might experience an increase in competitive service offerings. This finding is in line with Conroy et al.'s (2021) results with regard to population density. It is worth mentioning that population density can range dramatically, from around 1 to 1642 people per square mile, with the mean at 25. As for the Cox regression, neither variable could be included due to proportionality issues.

8.3. Product and service characteristics

Of the three Product and Service Characteristics—Speed, Discount, and Distance to BT Tower—all but the latter were significant in the KM Plots. Speed was also significant in the Cox regression, but Distance to BT Tower was not, and Discount could not be included. From a predictive point of view, Speed was the fifth most important variable (7.71%). The Cox regression results suggest that in going from low speed (<50 Mbps) to high speed (≥50 Mbps), there is a 40.5% decrease in churn risk. Our results regarding internet speed coincide with Bhattacharyya et al.'s (2021) research on mobile devices. Thus, offering higher speeds, such as 100 Mbps, would be especially attractive to remote workers who frequently engage in video conferencing and large file downloads. Furthermore, households with multiple users will increase the demand, as there is only a fixed capacity to be shared.

Price can also be a major issue, but BT's base prices are directly linked to internet speed; thus, only price discounts can be examined in this study. The RSF importance of discounts was large, at 11.8%, but this is primarily due to the $5 discount. BT is actually moving away from providing discounts due to the inconsistency of their offerings across zones. In fact, higher discounts, which are more likely for more expensive plans, do not have much more impact than the lower discounts. Obviously the overall price, which factors into a discount, can be important (Ganesh et al., 2000); however, our KM plots suggest that there is a minimal difference between a $5 discount and a larger one. This, coupled with correlation between speed and discount, indicates companies may need to go beyond discount to retain customers (Kumar et al., 2015).

Although the reliability of the internet connection was not directly captured by BT's CRM system, there is little evidence to show that reliability is a major issue for BT. Distance to the BT Tower was used as a proxy for reliability, as increased distance can lead to an attenuation of signal strength. Consequently, the distance between the BT tower and the household can indicate the likelihood of experiencing a connectivity issue. BT Tower Distance does not appear to be a substantive factor by examining the KM Plots, Cox regression, or RSF importance (2.4%). This suggests that either reliability is not an issue, as BT would claim, or that the distance to the tower is not a reliable proxy. In fact, several arguments against using the Distance to BT Tower as a proxy to reliability can be made. First, a repeater could be used to transmit and amplify the signal; however, that information is not readily known. Second, tower height can be a substantive factor for increasing reliability. To that point, BT tends to utilize taller towers whereas many of the other regional towers tend to be under 200 feet. Third, the reliability of the tower is quite dependent on the technology of the equipment used for transmission.

8.4. Competition

The presence of ISP competitors undoubtedly influences customer churn, yet understanding the type of connection technology is crucial in determining its significance. Prevalent technologies for rural environments, which are not omnipresent, include DSL, cable, fiber, and other FWA providers. However, DSL is no longer considered a strong competitor, and it only represents an RSF importance of less than one percent.

The growth in fiber deployment has been substantial in more recent years, and fiber is now available to over 50% of all (rural and urban) households (Cooper, 2023). Fiber is obviously the technology of choice for ISP providers since the technology is more future-proof and holds considerable advantages over cable (de Wit, 2022). Consequently, fiber represents a formidable challenge to FWA ISPs, especially given fiber's preference in governmental programs. Our results show that the presence of fiber is the most important variable for predicting churn. BT has countered this threat by the purchase of a fiber deployment company.

It is worth noting that fiber is an important component for even FWAs, as it provides the trunks for ISP connectivity (i.e., it's a hybrid solution). Despite large growth in fiber optic deployment, telecom providers still tend to "cherry pick" routes that are more profitable (Leighton & Clark, 2024). Consequently, to achieve rural connectivity for all will most likely involve the use of other types of connections such as FWA and satellite.

Cable, when available, usually offers a much faster connection than FWA, and is still a threat to BT, but its RSF important was only 1.6%. Cable was a significant factor in the KM Plot, but the survival curves were not that far apart. One possibility here is that Cable has been around for quite a few years, and customers who had this option, for the most part, have already defected if they wanted to.

Conceptually there is no doubt that another FWA provider can be a competitive threat to BT. The distance to a competitive tower was significant in the Cox regression but not in the KM plots. A more significant impact, which is not captured in our analysis but is observed in practice, is when a new FWA provider enters the BT service area. Usually, competitors offer a lower introductory price point, which temporarily increases BT customer churn. This observation is consistent with economic theories of market entry, where new entrants typically adopt aggressive pricing strategies to capture market share, thereby increasing churn for incumbent providers (Porter, 1980).

8.5. Local brand presence

BT has a lot of disparity when it comes to brand presence across its 13 sales zones. For the RSF model, the regrouped 5-level zone variable was the second most important variable. BT has the strongest brand reputation in Zones 1 and 2. These results would indicate that regional factors have a strong impact on customer retention. Such findings align with other research in mobile communications, where a strong brand image enhances a company's resilience against customer churn (Svendsen & Prebensen, 2013). Regarding brand image initiatives, it is worth emphasizing that BT engages in activities that go beyond traditional marketing initiatives, such as internet literacy training sessions held at local libraries via their BT University program.

9. Theoretical and managerial implications

This study had a substantive impact at BT, a sentiment echoed by Scott Feldt, BT's Executive Director of Public Affairs, who stated, "This churn study has been invaluable to us as we continue to identify and retain one of our most important assets—our customers." While BT values the ability to predict churn over time, their greater interest lies in understanding predictive importance, which has reaffirmed their decision to purchase a company for fiberoptic deployment. The results also highlight how different strategies across sales zones can influence customer retention, reflecting the localized challenges and opportunities faced by zone managers. These findings suggest that a combination of company-level strategy and zone-specific initiatives are key to improving customer loyalty and mitigating churn.

The findings also emphasize the need to focus on the technologies currently powering FWA services. While forward-looking advancements like 5G offer exciting possibilities, many FWA providers may rely on other wireless technologies, such as LTE, to deliver connectivity. These technologies, though limited in speed and latency compared to fiber, are still vital for bridging the rural connectivity gap in regions where fiber deployment remains economically or logistically unfeasible. Additionally, upgrades to the towers' transmission equipment can significantly improve service, but at the same time may require additional towers. Hybrid solutions that combine FWA for last-mile access with fiber backbones provide a practical and scalable approach, particularly in underserved rural areas. Recent shifts in National Telecommunications and Information Administration (NTIA) policy, including greater openness to FWA and satellite providers like Starlink under the BEAD initiative, underscore the growing recognition of the role of wireless in expanding broadband access. Strategic planning will be necessary to ensure these technologies remain sustainable and competitive while addressing market dynamics and evolving customer needs.

This study also has broader policy implications. The results reinforce the need for inclusive broadband programs that embrace mixed-technology approaches, recognizing the complementary roles of fiber, FWA, and satellite in connecting underserved areas. While fiber is ideal for high-density regions, hybrid solutions provide cost-effective alternatives for rural areas, where fiber deployment is difficult, ensuring no communities are left behind. Additionally, ISPs should be encouraged to adopt data-driven decision-making that also use customer experience measures, such as satisfaction levels. Finally, fostering digital literacy among underserved demographics, such as older customers, through partnerships with local providers or community organizations, could increase adoption and improve customer retention, as evidenced by BT University.

Another significant contribution of this study is its demonstration of how external data enhances churn analysis for small and medium-sized companies. By integrating data from external sources, such as third-party providers, the FCC, and the U.S. Census Bureau, BT was able to identify key drivers of churn that went beyond what internal company data could capture. External data provided insights into competition, population density, and customer demographics, enabling a more comprehensive understanding of the factors affecting churn. These findings underscore the importance of combining internal and external data sources to inform strategic decision-making and address customer retention challenges more effectively.

The results of this study yield three primary takeaways. First, price discounts alone are insufficient for mitigating churn, particularly for high-speed internet services. As previous research suggests, discounts have diminishing returns beyond a certain threshold, indicating that providers should explore alternative approaches to retain customers. Second, speed improvements remain critical, even with older FWA technologies. By investing in hybrid solutions that combine FWA with fiber, providers can meet the growing demand for higher speeds while maintaining cost-effectiveness. Third, older customers, who represent a significant segment of BT's rural customer base, require targeted engagement. Initiatives like BT University, should be enhanced to strengthen loyalty within this segment. While speed may not currently be as much of a priority for this group, changing habits and technological demands may shift this dynamic in the future.

Finally, churn studies like this one have applications that extend beyond customer retention. They can serve as the foundation for customer acquisition strategies, segmentation analyses, and lifetime value modeling. Expanding this research to incorporate additional factors, such as customer satisfaction, internet usage patterns, and service quality perceptions, would yield even deeper insights. BT is already considering additional research trajectories based on this study, ensuring that its approach to customer retention and acquisition continues to evolve with the demands of the telecommunications landscape.

10. Limitations and future research

This study focuses on an FWA ISP in the Midwest, predominantly operating in rural areas, and its findings should be cautiously generalized due to its specific context (i.e., sales zones) and its observational nature. Due to the limitations encountered, the landline competitiveness of an area was overstated due to an assumption implicit in the historical FCC data.

There were also some product and service characteristics not considered in this study due to lack of data availability and multicollinearity. For example, behavioral and satisfaction issues, such as internet usage, were not monitored, nor were attitudes captured for all the customers. Another example dealt with price and speed, which could not be considered together, as there is a direct relationship between the two. Additionally, pricing can be impacted by the use of hotspots, which is becoming a more viable competitor with 5G technology. The use of hot spots was not considered in this paper.

The intent of this research was to develop an explanatory model that contained most of the available variables set forth in our framework; however, this is not usually possible with the Cox regression model. Since proportionality assumptions cannot be ignored, the development of model(s) could become more involved, as this might require either time-dependent variables, the exclusion of variables, or the creation of multiple models. The importance of many of these omitted variables can be ascertained using an RSF.

This research has significantly advanced the understanding of how data analytics can be leveraged in churn studies involving FWA. The methodology outlined in this paper is particularly valuable for FWA ISPs, offering a framework for analysis that can be adapted to their specific needs. The multi-method approach used in this study provides a thorough examination of the available variables (i.e., variables that may need to be excluded in some approaches can be used in others).

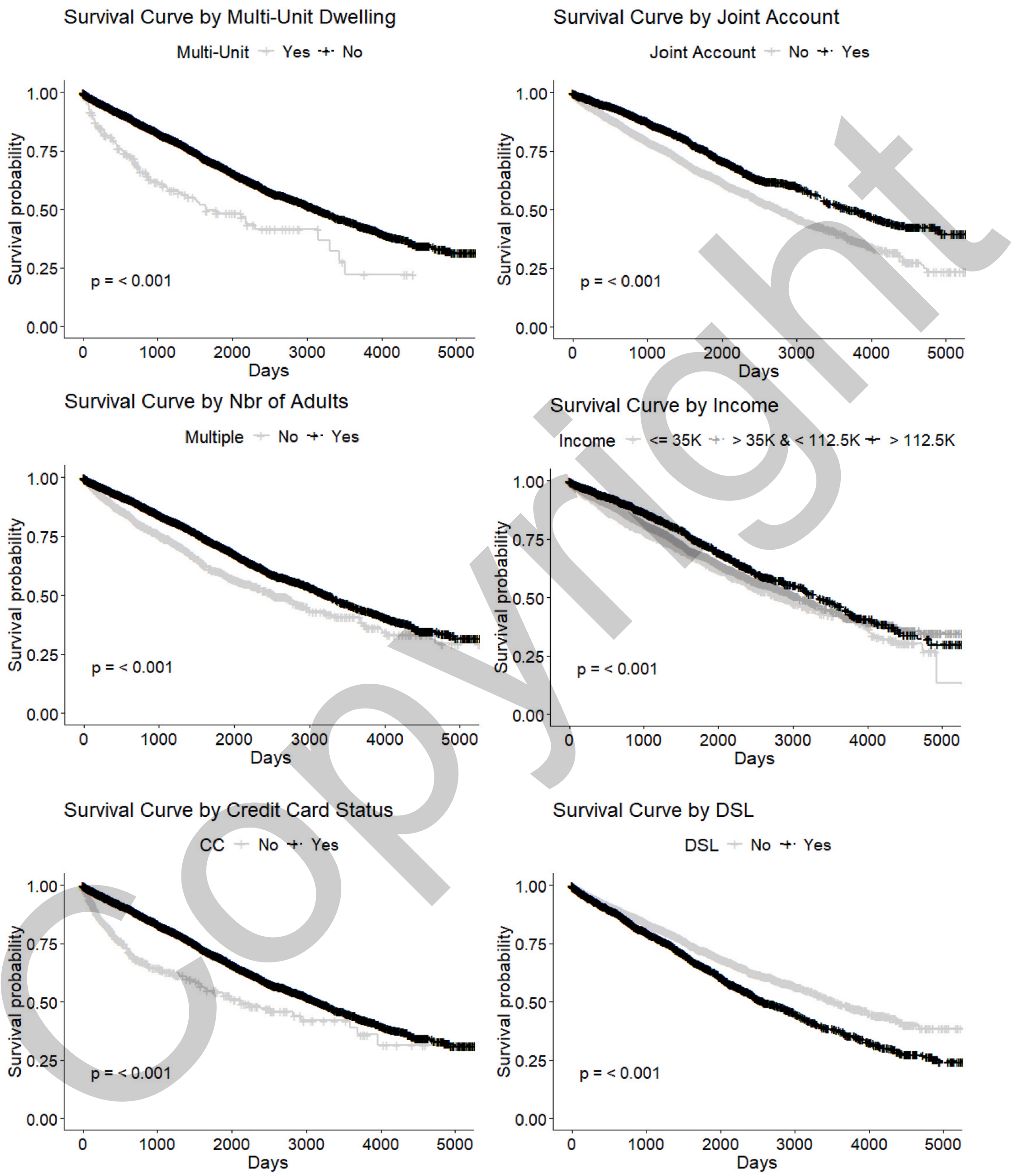
There are several avenues to expand on for future research on FWAs. First and foremost, other churn studies could replicate the framework presented in this study to assure generalizability of results. Second, this study was not able to investigate all of the subcategories, such as internet usage, presented in the framework; therefore, a more expansive study would be advantageous. Third, being able to stratify the analysis by subcategories, such as price, speed, and various levels of service quality within 5G offerings, could provide additional insights.

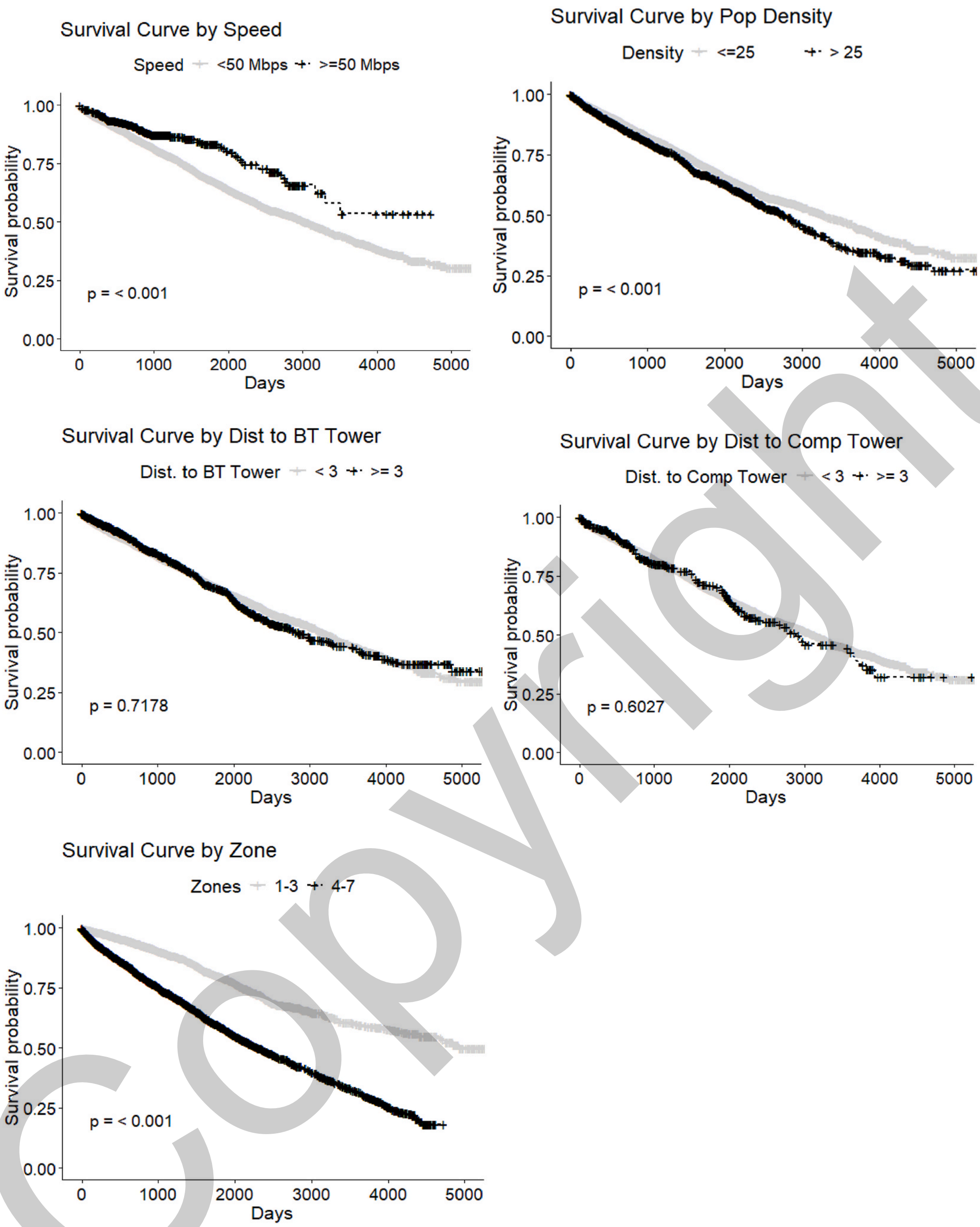
CRediT authorship contribution statement

Kurt A. Pflughoeft: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nikolaus T. Butz:** Writing – review & editing, Writing – original draft, Investigation. **Alick Corbley:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Appendix A

Kaplan-Meier Plots.


Survival Curve by Multi-Unit Dwelling
Multi-Unit Yes No
p = < 0.001
Survival Curve by Joint Account
Joint Account No Yes
p = < 0.001
Survival Curve by Nbr of Adults
Multiple No Yes
p = < 0.001
Survival Curve by Income
Income <= 35K > 35K & < 112.5K > 112.5K
p = < 0.001
Survival Curve by Credit Card Status
CC No Yes
p = < 0.001
Survival Curve by DSL
DSL No Yes
p = < 0.001
Survival probability
1.00
0.75
0.50
0.25
0.00
Days
0
1000
2000
3000
4000
5000



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